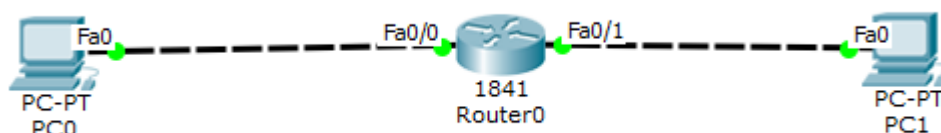
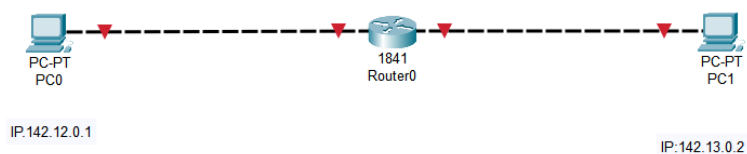


Date:

EXPERIMENT NO. 5**AIM: Understand and identify Layer-3 functionality.****Exercise-1:**

Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your knowledge.



ARP table entry of PC0 after ping

142.13.0.1	0060.5CCD.2BC2
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ARP table entry of PC1 after ping

142.12.0.1	0001.6444.C8E0
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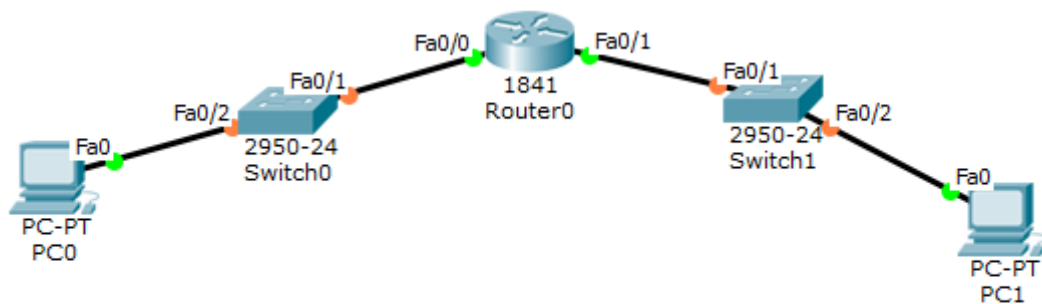
ARP table entry of Router0:

142.12.0.1	0001.6444.C8E0	FastEthernet0/0
142.12.0.2	0001.64B9.3C01	FastEthernet0/0
142.13.0.1	0001.64B9.3C02	FastEthernet0/1
142.13.0.2	0060.5CCD.2BC2	FastEthernet0/1

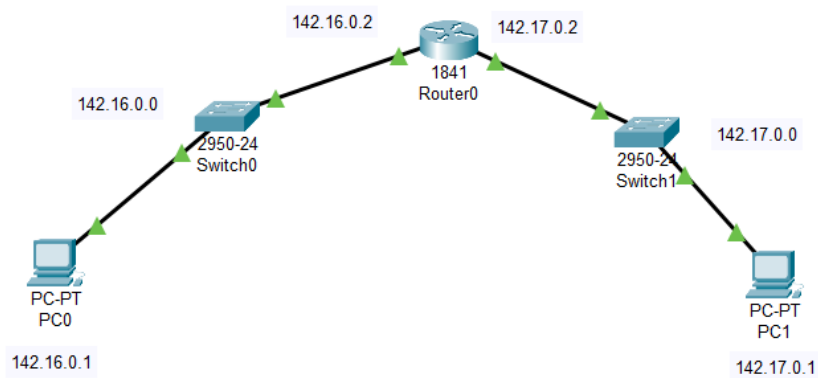
Routing table of Router0:

Type	Destination Network ID	Next Hop IP address	Hop Count along a path
C	142.12.0.0/16	--	--
C	142.13.0.0/16	--	--

Exercise-2



Redraw above diagram which includes IP address and MAC address. Take IP address and MAC address as per your knowledge.



ARP table entry of PC0 after ping

142.16.0.2	0009.7CA9:B301
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ARP table entry of PC1 after ping

142.17.0.2	0009.7CA9:B302
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Switch table entry of Switch0:

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Switch table entry of Switch1:

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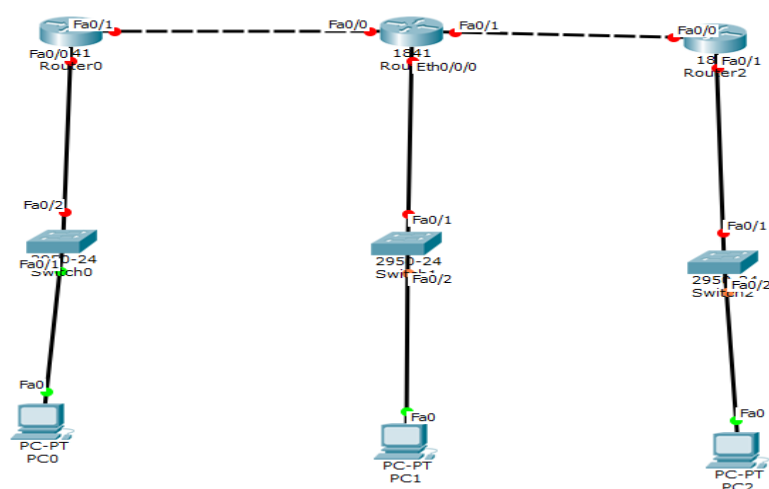
ARP table entry of Router0:

IP address	MAC Address	Interface
142.16.0.1	0005.5ED7.C653	FastEthernet0/0
142.16.0.2	0009.7CA9:B301	FastEthernet0/0
142.17.0.1	00D0:58C9:ED5D	FastEthernet0/1
142.17.0.2	0009.7CA9:B302	FastEthernet0/1

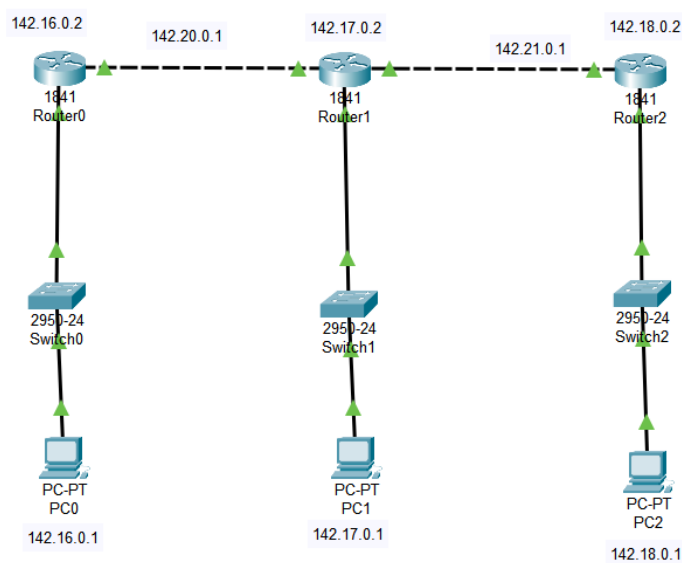
Routing table of Router0:

Type	Destination Network ID	Next Hop IP address	Hop Count along a path
C	142.16.0.0/16	--	--
C	142.17.0.0/16	--	--

Exercise-3



Redraw the above diagram which includes the IP address and MAC address. Take the IP address and MAC address as per your knowledge.



ARP table entry of PC0

142.16.0.2	0002.4A17.CD01
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ARP table entry of PC1

142.17.0.2	0009.0CCC.3701
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ARP table entry of PC2

142.17.0.2	0009.7C97.0601
142.18.0.2	0009.7C97.0601

Switch table entry of Switch0:

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Switch table entry of Switch1:

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Switch table entry of Switch2:

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ARP table entry of Router0:

IP address	MAC Address	Interface
142.16.0.1	00D0.D3EB.19EE	FastEthernet0/0
142.16.0.2	0002.4A17.CD01	FastEthernet0/0
142.20.0.1	0002.4A17.CD01	FastEthernet0/1
142.20.0.2	0090.0CCC.3702	FastEthernet0/1

ARP table entry of Router1:

IP address	MAC Address	Interface
142.17.0.1	0009.0C52.37D7	FastEthernet0/0
142.17.0.2	0090.0CCC.3701	FastEthernet0/0
142.20.0.1	0002.4A17.CD01	FastEthernet0/1
142.20.0.2	0090.0CCC.3702	FastEthernet0/1
142.20.0.1	0009.0C90.3C41	FastEthernet0/1/0

ARP table entry of Router2:

IP address	MAC Address	Interface
142.18.0.1	0060.5CCC.97CD	FastEthernet0/0
142.18.0.2	0009.7C97.0601	FastEthernet0/0
142.20.0.1	0002.4A17.CD01	FastEthernet0/1
142.20.0.2	0090.0CCC.3702	FastEthernet0/1

Routing table of Router0:

Type	Destination Network ID	Next Hop IP address	Hop Count along a path
C	142.16.0.0/16	--	--
S	142.17.0.0/16	142.20.0.2	--
S	142.18.0.0/16	142.20.0.2	
C	142.20.0.0/16	--	--

Routing table of Router1:

Type	Destination Network ID	Next Hop IP address	Hop Count along a path
S	142.16.0.0/16	155.20.0.1	--
C	142.17.0.0/16	--	--
S	142.18.0.0/16	155.21.0.2	--
C	142.20.0.0/16	--	--
C	142.21.0.0/16	--	--

Routing table of Router2:

Type	Destination Network ID	Next Hop IP address	Hop Count along a path
S	142.16.0.0/16	142.20.0.1	--
S	142.17.0.0/16	142.20.0.1	--
C	142.18.0.0/16	--	--
C	142.21.0.0/16	--	--

CONCLUSIONS:

Pre-Lab Assignment Questions

1. How do DNS work?

2. What is the significance of the routing table?

3. What is the next hop IP address?

4. What is the significance of routing protocols?

Post-Lab Assignment Questions**Exercise 1**

Which are the following statements correct? Also, justify each statement.

1. Does PC0 have an Ethernet Card?
2. PC1 is having Ethernet Card.
3. Router0 has two NIC cards.
4. Router 0 has two Mac addresses.
5. Router0 has TWO IP address.
6. MAC address pair on link 0(between PC0 and Router0) is different than MAC address pair in link 1(between Router0 and PC1) for message transfer.

7. Router0 has a switching table
8. The speed of Link 0 is 10 Mbps.
9. The speed of Link1 is 100 Mbps.
10. Router0 makes decisions based on MAC address.
11. PC0 and PC1 can communicate because they have the same Ethernet cards i.e. Fa0.
12. PC1 and Router0 cannot communicate as they have different Ethernet cards.

Exercise 2

Which are the following statements correct? Also, justify each statement.

1. Switch0 contains MAC address of PC1 in their switching table.
2. Switch1 contains MAC address of PC1 in their switching table.
3. Any computer or device can be a Router with two NIC cards.
4. Switch0 and Switch1 may take decisions based on IP address.
5. Router0 works at layer 3, while switches work at layer 2.
6. By default, the Network ID of NIC cards are routing table entries.
7. MAC address pair on link 0(between PC0 and Router0) is different than MAC address pair in link 1(between Router0 and PC1) for message transfer.